

Thermodynamic Modeling: L-aspartic acid monosodium salt monohydrate

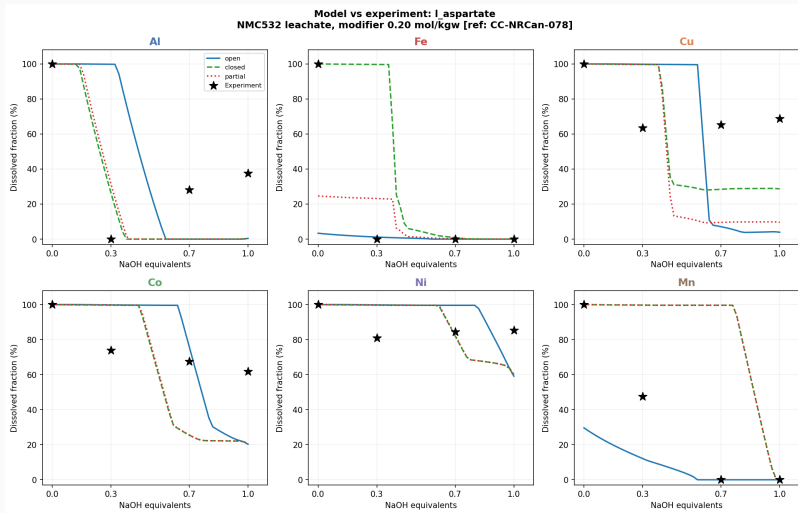
NMC532 Leachate, Modifier 0.20 mol/kgw

MDPSA Pipeline

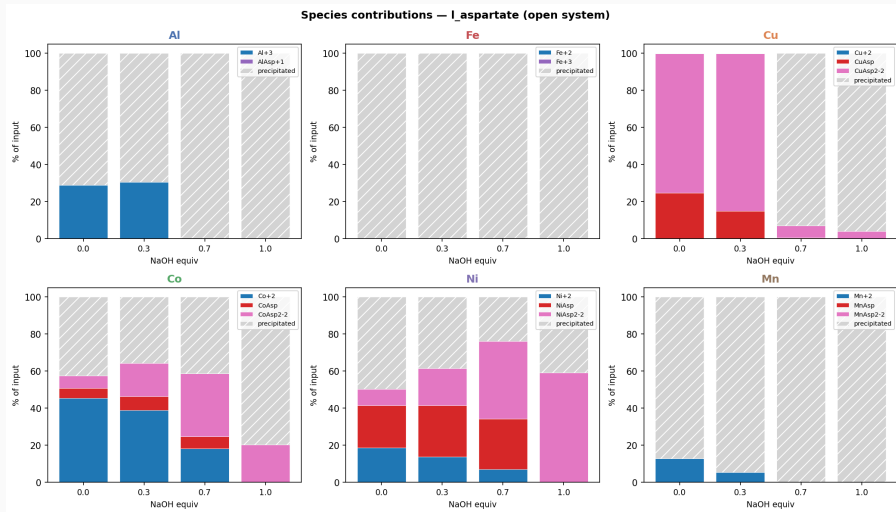
2026-07-22

NRCan CC-NRCan-078

Model vs. Experiment



Species Contributions (Open System)



Selectivity Scorecard

Metal	Role	@ 0.3 eq	@ 1.0 eq	RMSE (pp)	Verdict
Cu	Target	100%	4%	47.2	Partial
Co	Target	100%	20%	24.8	Partial
Ni	Target	100%	59%	17.8	Protected
Mn	Target	12%	0%	39.4	Lost
Al	Impurity	100%	0%	55.1	Partial
Fe	Impurity	1%	0%	48.3	Removed

RMSE: root-mean-square error in percentage points vs. experiment (CC-NRCan-078).

Protected/Removed: model matches experiment at all comparison equiv points.

Thermodynamic Parameters: L-aspartic acid monosodium salt monohydrate

Species	log K
CuAsp	8.62
CuAsp2-2	14.00
CoAsp	5.49
CoAsp2-2	10.50
NiAsp	6.52
NiAsp2-2	11.00
MnAsp	3.20
MnAsp2-2	2.50
AlAsp+1	3.50

- Database: sit.dat (SIT activity model)
- Modifier dose: 0.20 mol/kgw
- Temperature: 25°C, I = 0 (NIST reference)
- Equilibrium phases: Gibbsite, Ferrihydrite(am), $\text{Cu(OH)}_2(\text{s})$, $\text{Co(OH)}_2(\text{s})$, $\text{Ni(OH)}_2(\text{s})$, MnOOH
- Fe complexation suppressed (kinetic)
- Scenarios: open / closed / partial CO_2

Source: NIST Standard Reference Database 46 Version 8.0. Values adjusted for ionic-strength effects at leachate conditions.